1 Abstract

In this paper, we developed and compared nondestructive techniques for the purpose of determining the ripeness state (classified as either reached ripeness or not – 2 states) of local berry fruit based on 5 machine learning methods: Decision Learning Tree, Quadratic discriminant, Support vector machine, Ensemble learning and kNN (k – Nearest Neighbors). Infrared thermal images of freshly plucked berries (ripen and unripen) were taken at two different instances. The relative size of higher temperature inner cores and its average temperature was recorded. These served as predictors or features used in each model.

Keywords: non-destructive testing, fruit ripeness, Machine learning classifiers, MajorAxisLength, Area, Inner Core, infrared

2 Introduction

One of the most important objectives in the food industry is that of achieving a uniform quality both of raw materials and of the final product. One of the main concerns of the fruit industry is the systematic determination of fruit ripeness under harvest and post-harvest conditions, because a variability in ripeness is perceived by consumers as a lack of quality.

Most of the traditional methods that have been used to assess fruit ripeness are destructive and thus cannot be so readily applied. Some of these methods rely on the measurement of fruit firmness, which is correlated to ripeness, by using a penetrometer [1] or an impact force [2]. Other strategies include measuring levels of chemical species and parameters that are correlated to ripeness such as pH, titratable acidity [3], soluble-solids contents (i.e. sugars) [4] and ethylene contents [5]. The measurement of these chemical species and parameters generally requires once again the destruction of the fruit together with complex analytical techniques, such as gas–liquid chromatography (ethylene) and titration (acidity) [6]. More recently, non-destructive methods of determining fruit ripeness have been proposed. These methods include

(i) nuclear magnetic resonance (NMR) [7] and proton magnetic resonance (PMR) [8] to determine levels of soluble solids,
(ii) vision systems to determine skin colour of fruit (colour has been shown to be correlated to sugar contents) [9] and
(iii) acoustic methods to determine fruit firmness [10]. However, there are problems inherent to these approaches. NMR and PMR techniques rely upon expensive equipment. Acoustic methods present the problem of how to reproducibly couple the acoustic impedances of the emitter and the fruit to be tested. Fruit colour is, in some cases, only weakly correlated to fruit ripeness.

An attractive and alternative strategy for determining the state of ripeness of fruit is to make use of trained machine learning classifiers on computationally efficient systems. Furthermore, these classification models were able correctly classify response vectors of with accuracies of 58.3% by Complex tree, 66.7% by Quadratic Discriminant, 91.7% by Quadratic SVM, 83.3% by Fine kNN and 91.7% using Ensemble bagged tree model.

3 Methodology

3.1 Materials

The berries were plucked freshly from trees within the institute (Indian Institute of Technology, Kharagpur). They were then gently brushed off to remove dust or any other foreign particles which may interfere with the fruit's thermal signature. Two such batches of berries were supplied for the same. First batch (plucked 2nd March 2018) to serve as a 'training set', the accuracy of which could later be measured by classifications made using 'test set' – the second batch (plucked 1st May 2018).

The infrared spectrum thermal images were taken using a commercial infrared camera with a temperature resolution of 0.1 K. The temperature of various parts of fruit would be available via these images.

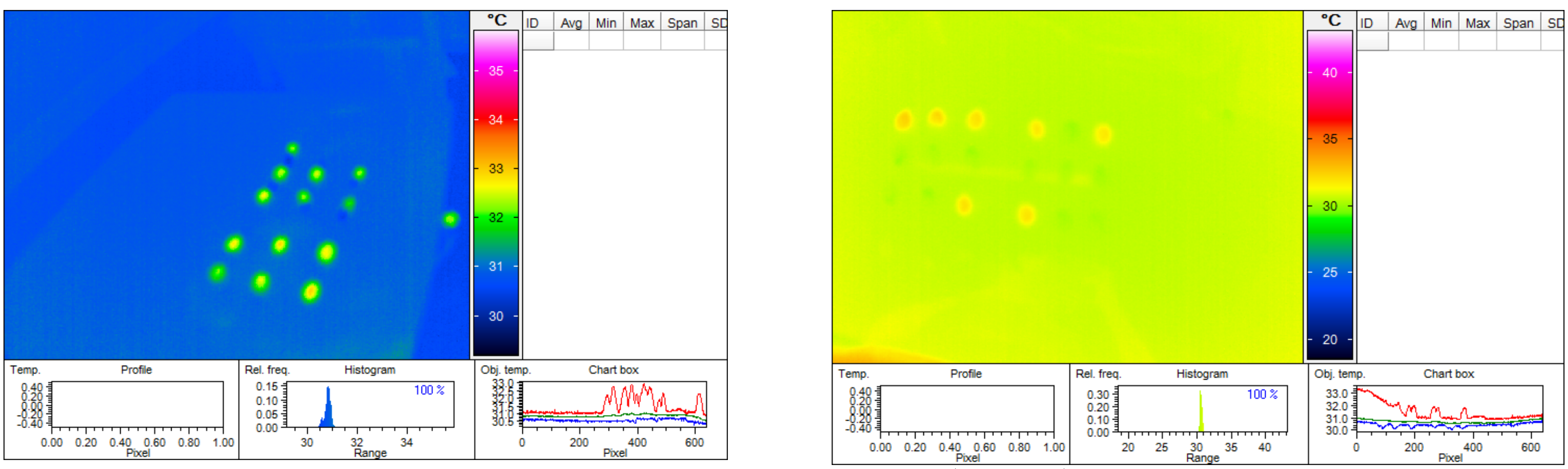


Fig. 1: Infrared thermal images of first batch/training set (left side) and of second batch/test set (right side)

These images were then extensively processed and analyzed on MATLAB® (version R2017a). Principal features from it which were employed in the techniques were

- Image Processing toolbox
  Image Processing Toolbox provides a comprehensive set of reference-standard algorithms and workflow apps for image processing, analysis, visualization, and algorithm development. We can perform image segmentation, image enhancement, noise reduction, geometric transformations, image registration, and 3D image processing.
- Classification Learner App
  The Classification Learner app trains models to classify data. We explored our data, selected features, specified validation schemes, trained models, and assessed the

results. Automated training was done to search for the best classification model type, including decision trees, discriminant analysis, support vector machines, logistic regression, nearest neighbors, and ensemble classification.

Supervised machine learning was performed by supplying a known set of input data (observations or examples) and known responses to the data (e.g., labels or classes). We used this data to train a model that generated predictions for the response to new data. To use the model with new data the model was exported to the workspace.

3.2 Test procedures

The sampling of data from train-set berries involved: -

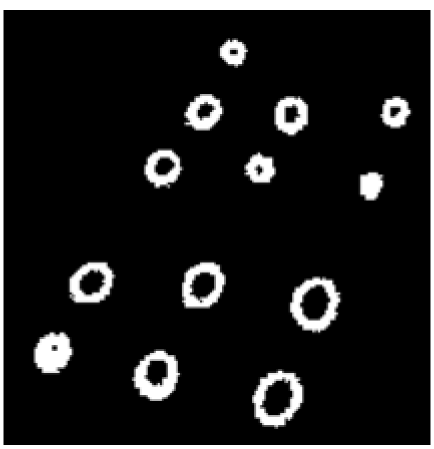
(i) taking Infrared thermal images of the berry

(ii) extracting 3 features: average core temperature, relative core size(Area and $MajorAxisLength$).

(iii) Relative core size stands for

a) $Area\ of\ berry\ inner\ core \div Area\ of\ berry$

b) $MajorAxisLength\ of\ berry\ inner\ core \div MajorAxisLength\ of\ berry$



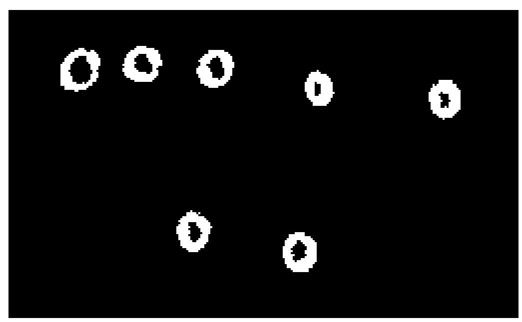


Fig. 2: Processed images of first batch/training set (left side) and of second batch/test set (right side)

(iv) Training the supervised machine models on classifier learner app using this data.

(v) Comparing these models w.r.t the highest prediction accuracy. So, among each of the five model type (tree, SVM, kNN, discriminant and ensemble) one can be ‘extracted’ from classifier learner app and later used on test.

3.3 The use of high temperature inner core of berries as a class reference

Thermal treatment methods of fruits using hot water, vapor or hot air have been investigated extensively as alternatives to methyl bromide fumigation for fruit disinfestation [11].

Varying degrees of efficacy have been reported using different thermal treatments alone or in combination with cold or controlled atmosphere (CA) storage conditions [12]. *Since insects may stay in the center of the fruit, the thermal energy needs to be delivered to that location.*

A common difficulty with hot air or water heating methods is the slow rate of heat transfer resulting in hours of treatment time [13]. Reported heating times for fruit center to reach the desired maximum temperatures range from 23 min for cherries to 6 h for apples. Thermal energy delivered to the interior of the fruits was significantly influenced by fruit size, heating medium temperature and heating methods. Slow conductive heat transfer often led to heating times in excess of 60 min [14].

Thus the temperature difference between the surface and core of the fruit exists due to different thermal diffusivities [14] and also shows dependence on fruit size, heating medium temperature/speed and heating methods.

4 Data Analysis

4.1 Observations

Table 1: Observations of three features (area, MajorAxisLength and Core temperature) for Training batch (taken at 2nd March 2018)

	Ripen			Unripen		
	Ratio		Temperature	Ratio		Temperature
S.L. NO.	Area	MajorAxisLength	(in °C)	Area	MajorAxisLength	(in °C)
1	0.022099448	0.135933109	32.1	0.344262295	0.494994449	32.6
2	0.474025974	0.505642361	32.6	0.232	0.450669881	32.4
3	0.338308458	0.47723796	32.5	0.077922078	0.314827697	32.2
4	0.479041916	0.508861897	32.7	0.072916667	0.313706786	32.2
5	0.626213592	0.575304722	32.8	0.236220472	0.439575196	32.5
6	0.592039801	0.567788364	32.6	0	0	31.9
7				0.144444444	0.384933566	32.2

Table 2: Observations of three features (area, MajorAxisLength and Core temperature) for Test batch (taken at 1st May 2018)

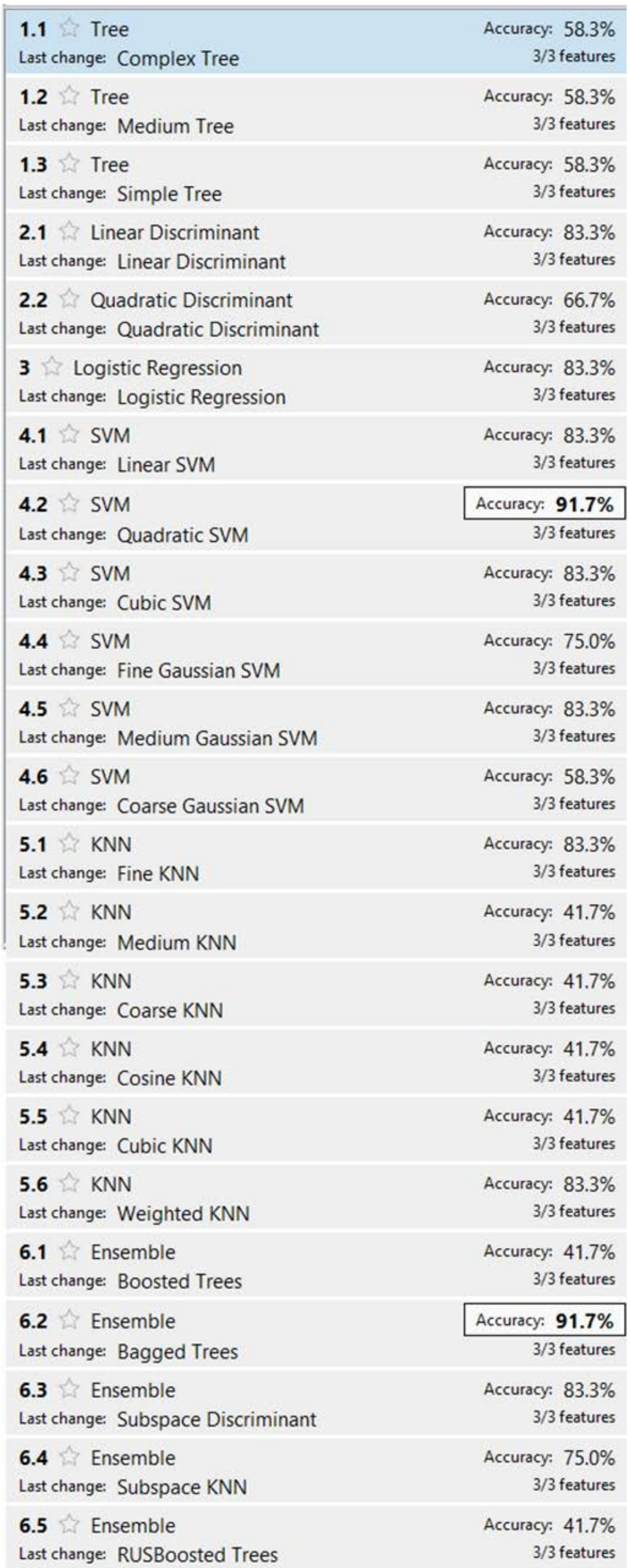
	Ripen			Unripen		
	Ratio		Temperature*	Ratio		Temperature*
S.L. NO.	Area	MajorAxisLength	(in °C)	Area	MajorAxisLength	(in °C)
1	0.444794953	0.692297	32.7	0.199275362	0.50709455	32.1
2	0.352941176	0.626633	32.6	0.092307692	0.403384465	31.9
3	0.442798354	0.566756	32.3	0.116935484	0.430417467	31.9
4	0.475471698	0.579162	32.3			

* Note that in Table 2, all temperatures were increased by 0.2 °C to allow test batch emulate train batch observation conditions and overall better fitting into the classes by classifier models. This was done to requite the difference in environmental temperature differences in train and test environment.

4.2 Training of Supervised Machine Learning Models

As, mentioned earlier, MATLAB®'s 'Classification Learner App' was used for the purpose of training the models using training batch.

Three features used were Relative(ratio) of MajorAxisLength, Area and Core Temperature. PCA was not employed.



Model	Type	Last change	Accuracy	Features
1.1	Tree	Complex Tree	58.3%	3/3 features
1.2	Tree	Medium Tree	58.3%	3/3 features
1.3	Tree	Simple Tree	58.3%	3/3 features
2.1	Linear Discriminant	Linear Discriminant	83.3%	3/3 features
2.2	Quadratic Discriminant	Quadratic Discriminant	66.7%	3/3 features
3	Logistic Regression	Logistic Regression	83.3%	3/3 features
4.1	SVM	Linear SVM	83.3%	3/3 features
4.2	SVM	Quadratic SVM	**91.7%**	3/3 features
4.3	SVM	Cubic SVM	83.3%	3/3 features
4.4	SVM	Fine Gaussian SVM	75.0%	3/3 features
4.5	SVM	Medium Gaussian SVM	83.3%	3/3 features
4.6	SVM	Coarse Gaussian SVM	58.3%	3/3 features
5.1	KNN	Fine KNN	83.3%	3/3 features
5.2	KNN	Medium KNN	41.7%	3/3 features
5.3	KNN	Coarse KNN	41.7%	3/3 features
5.4	KNN	Cosine KNN	41.7%	3/3 features
5.5	KNN	Cubic KNN	41.7%	3/3 features
5.6	KNN	Weighted KNN	83.3%	3/3 features
6.1	Ensemble	Boosted Trees	41.7%	3/3 features
6.2	Ensemble	Bagged Trees	**91.7%**	3/3 features
6.3	Ensemble	Subspace Discriminant	83.3%	3/3 features
6.4	Ensemble	Subspace KNN	75.0%	3/3 features
6.5	Ensemble	RUSBoosted Trees	41.7%	3/3 features

Fig. 3: Accuracy of predicting response vector of different types of Supervised machine learning classifiers:

1. Tree
2. Discriminant
3. Logistic Regression
4. Support vector machine
5. kNN
6. Ensemble

For each type of learning algorithm, the model with highest accuracy was chosen for final comparison. The select models were extracted from the app.

The extracted models(structures) was then used to assess prediction of response on new data set or test batch.

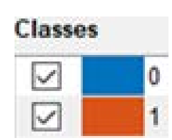
To make predictions simply the following line of code had to be used:

$$yfit = trainedModel.predictFcn(T);$$

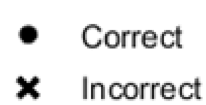
- On table T containing the test data
- And 'trainedModel' as the classification model

The conventions to follow in observations henceforth:

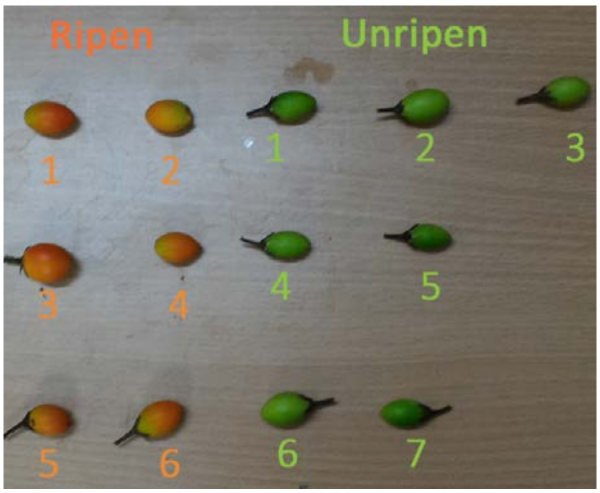
- Response of 1 stands for Ripen state and 0 for unripen state.



- In Scatter plot,



- Numbering scheme of the berries was followed throughout the experimentaion. As can be seen in Fig.



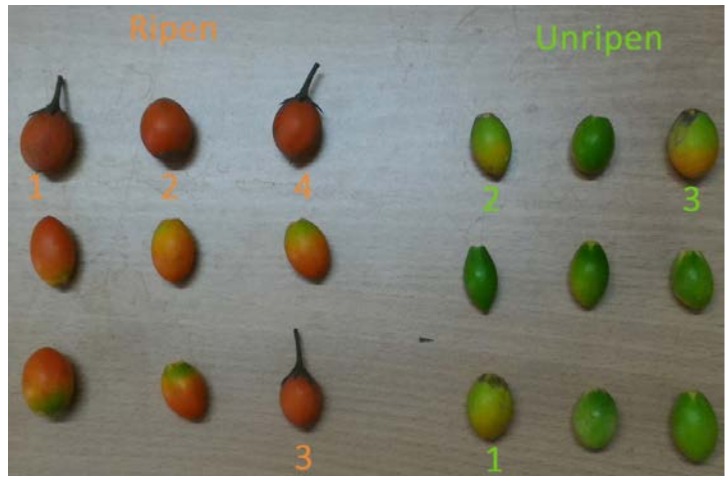
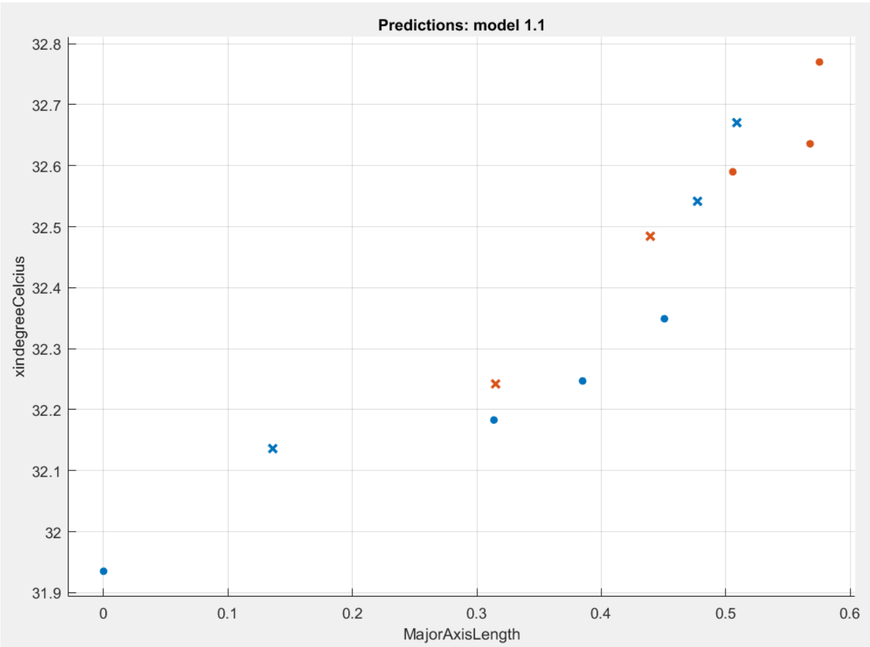


Fig. 4: Images of first batch/training set (left side) and of second batch/test set (right side), after numbering

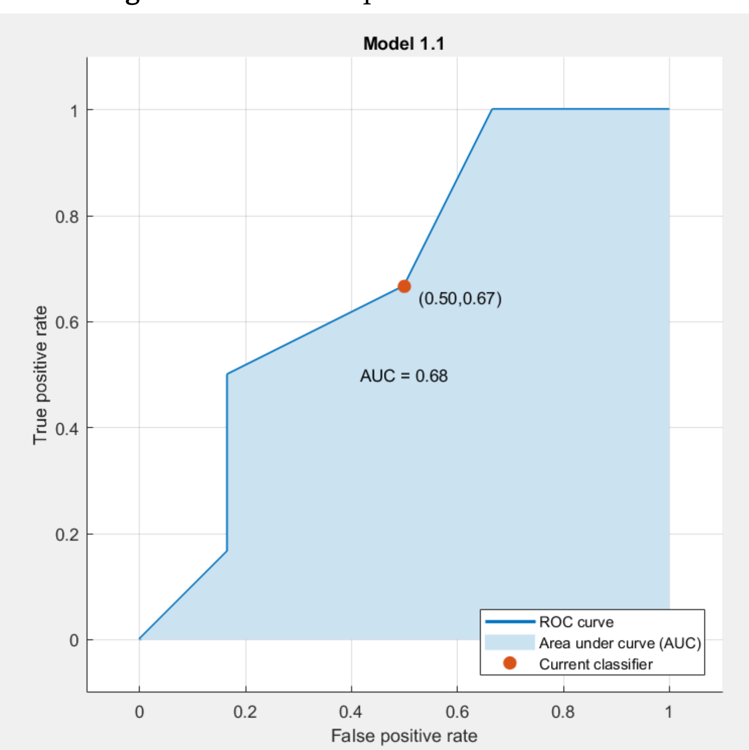
4.2.1 Complex Tree

Fig.5: Scatter Plot for Complex Tree model



A tree can be "learned by splitting the source set into subsets based on an attribute value test. This process is repeated on each derived subset in a recursive manner called recursive partitioning. The recursion is completed when the subset at a node has all the same value of the target variable, or when splitting no longer adds value to the predictions. This process of top-down induction of decision trees (TDIDT) [15] is an example of a greedy algorithm, and it is by far the most common strategy for learning decision trees from data.

Fig.6: ROC for Complex Tree model

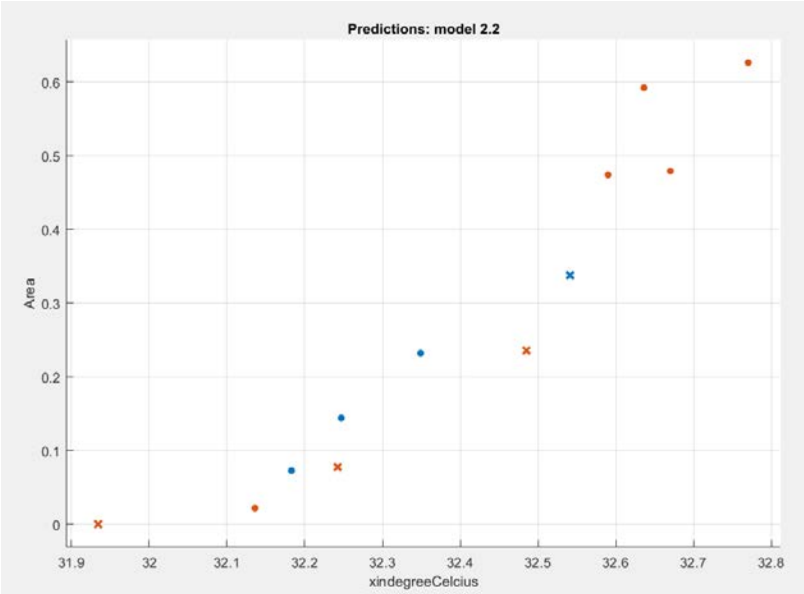


AUC of just 0.66 makes it seem like a random guesser (AUC = 0.5)

Accuracy of predicting response vector was also very low at only 58.3%.

4.2.2 Quadratic Discriminant

Fig. 7: Scatter Plot for Quadratic Discriminant model



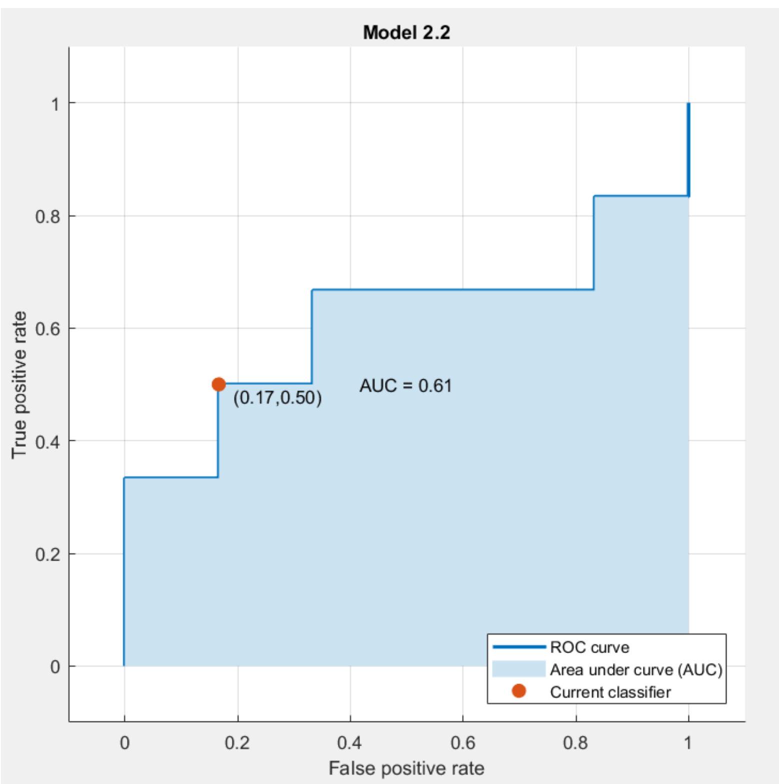
Quadratic discriminant analysis (QDA) is closely related to linear discriminant analysis (LDA), where it is assumed that the measurements from each class are normally distributed. Unlike LDA however, in QDA there is no assumption that the covariance of each of the classes is identical. When the normality assumption is true, the best possible test for the hypothesis that a given measurement is from a given class is the likelihood ratio test. Suppose there are only two groups, (so $y \in \{0,1\}$), and the means of each class are defined to be $\mu_{y=0}, \mu_{y=1}$ and the covariances are defined as $\Sigma_{y=0}, \Sigma_{y=1}$. Then the likelihood ratio will be given by

$$\text{Likelihood ratio} = \frac{\sqrt{|2\pi\Sigma_{y=1}|}^{-1} \exp\left(-\frac{1}{2}(x-\mu_{y=1})^T \Sigma_{y=1}^{-1}(x-\mu_{y=1})\right)}{\sqrt{|2\pi\Sigma_{y=0}|}^{-1} \exp\left(-\frac{1}{2}(x-\mu_{y=0})^T \Sigma_{y=0}^{-1}(x-\mu_{y=0})\right)} < t$$

for some threshold t.

After some rearrangement, it can be shown that the resulting separating surface between the classes is a quadratic. The sample estimates of the mean vector and variance-covariance matrices will substitute the population quantities in this formula [16].

Fig. 8: ROC for Quadratic Discriminant model

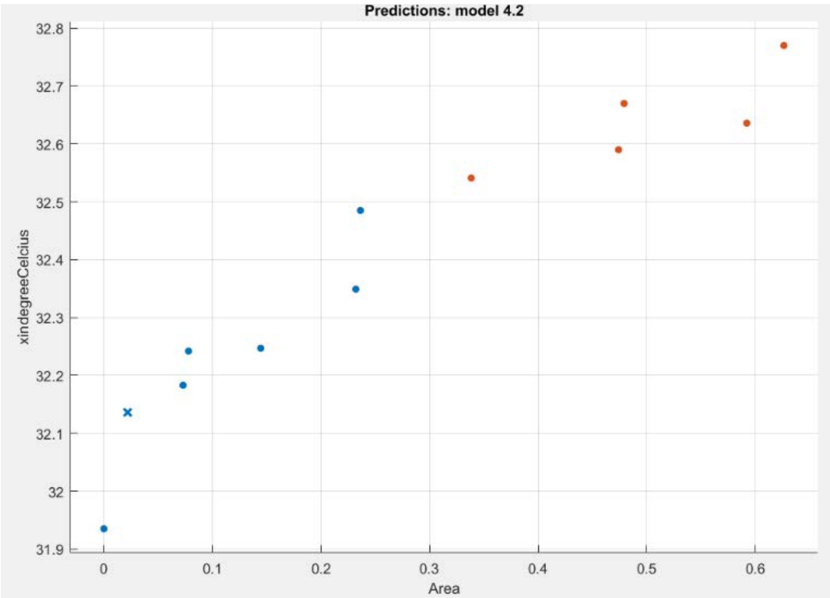


AUC of just 0.61 makes it seem like a random guesser (AUC = 0.5)

Accuracy of predicting response vector was also very low at only 66.7%.

4.2.3 Quadratic Support Vector Machine

Fig. 9: Scatter Plot for Quadratic SVM model

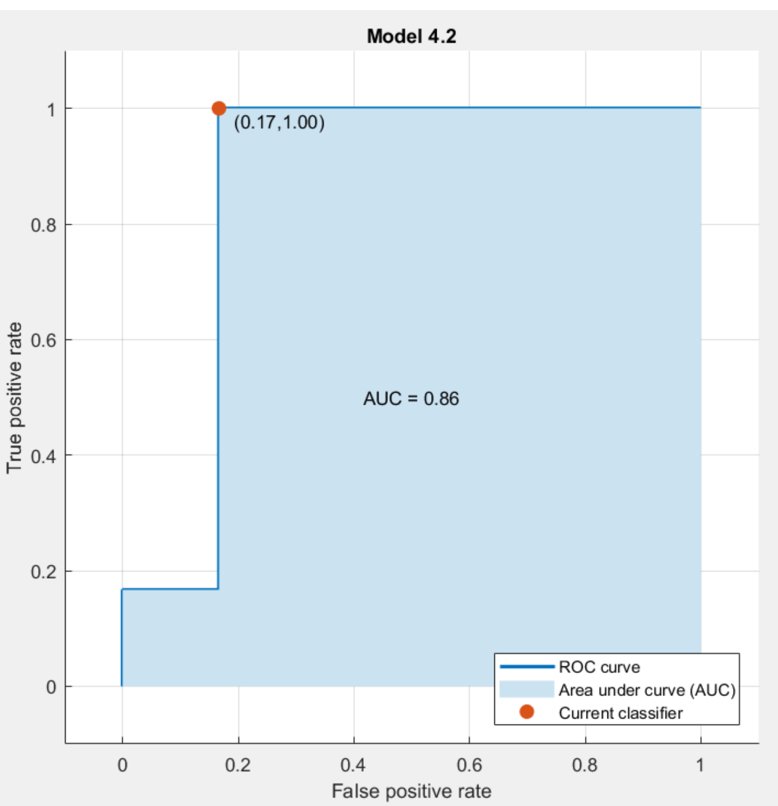


[17]

The SVM optimization problem can be stated as follows: Maximize the geometrical margin subject to all the training data with a functional margin greater than a constant. The functional margin is equal to $W^T X + b$ which is the equation of the hyper-plane used for linear separation. The geometrical margin is equal to $\frac{1}{\|W\|}$. And the constant in this case is equal to one.

To separate the data non-linearly, a dual optimization form and the Kernel trick is used. A quadratic decision function that is capable of separating the data non-linearly. The geometrical margin equal to the inverse of the norm of the gradient of the decision function.

Fig. 10: ROC for Quadratic SVM model

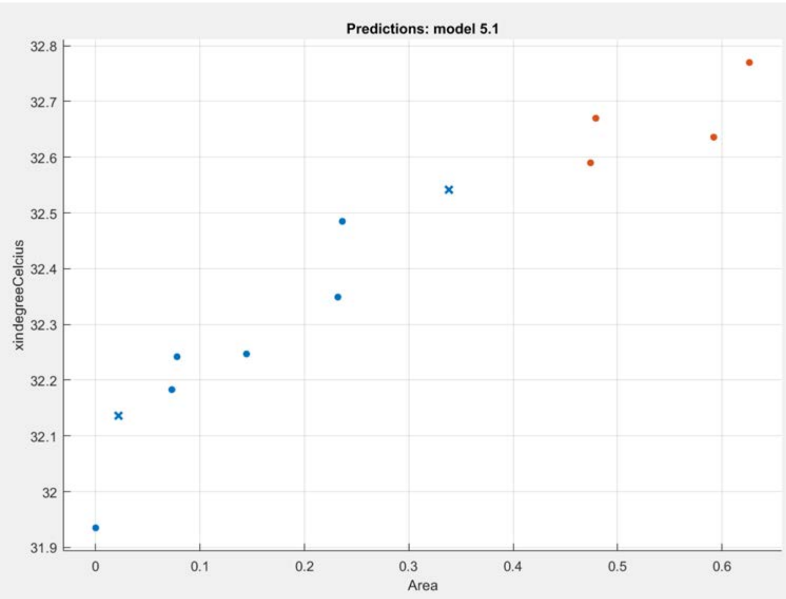


AUC of 0.86 is peerless for our purpose

Accuracy of predicting response vector was the highest amongst all supervised classifiers at 91.7%.

4.2.4 KNN (Fine) model

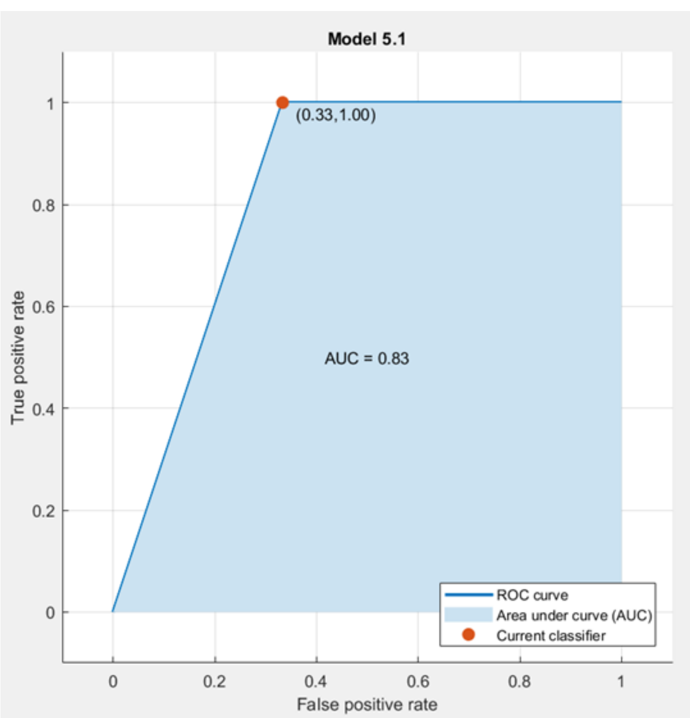
Fig. 11: Scatter Plot for KNN (Fine) model



In pattern recognition, the k-nearest neighbors algorithm (k-NN) is a non-parametric method used for classification and regression [17]. In both cases, the input consists of the k closest training examples in the feature space. The output depends on whether k-NN is used for classification or regression. In k-NN classification, the output is a class membership. An object is classified by a majority vote of its neighbors, with the object being assigned to the class most common among its k nearest neighbors (k is a positive integer, typically small). If k = 1, then the object is simply assigned to the class of that single nearest neighbor.

Fine kNN finely identify the nearest neighbors, which is very useful for eliminating the side-effect on classification of the test sample of the training samples that are far from the test sample and for enhancing the positive effect on classification of the test sample of the training samples that are close to the test sample

Fig. 12: ROC for KNN (Fine) model

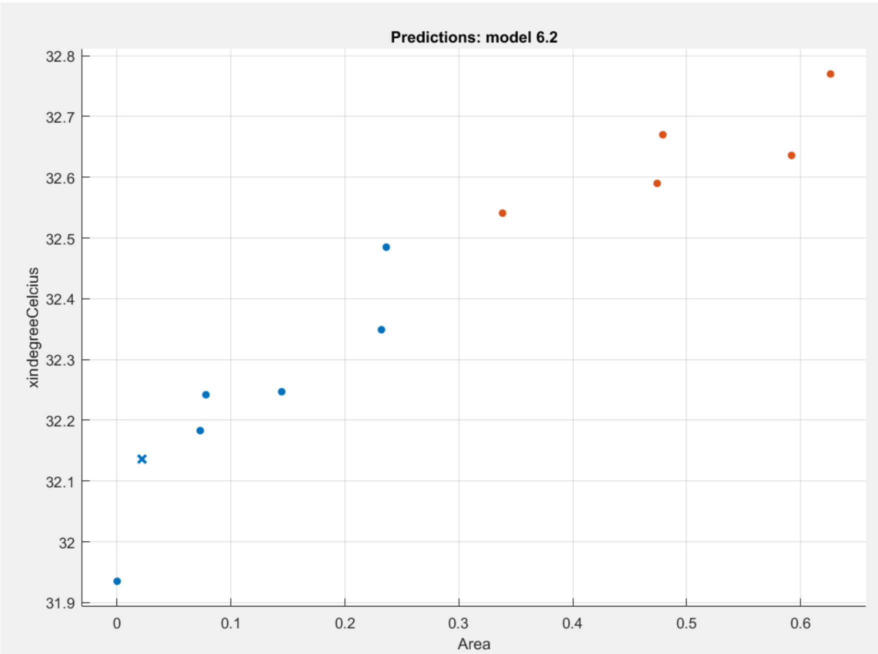


AUC of 0.83 is appreciable for our purpose

Accuracy of predicting response vector was notable 83.3%.

4.2.5 Ensemble: Bagged Trees model

Fig. 13: Scatter Plot for Ensemble: Bagged Trees model



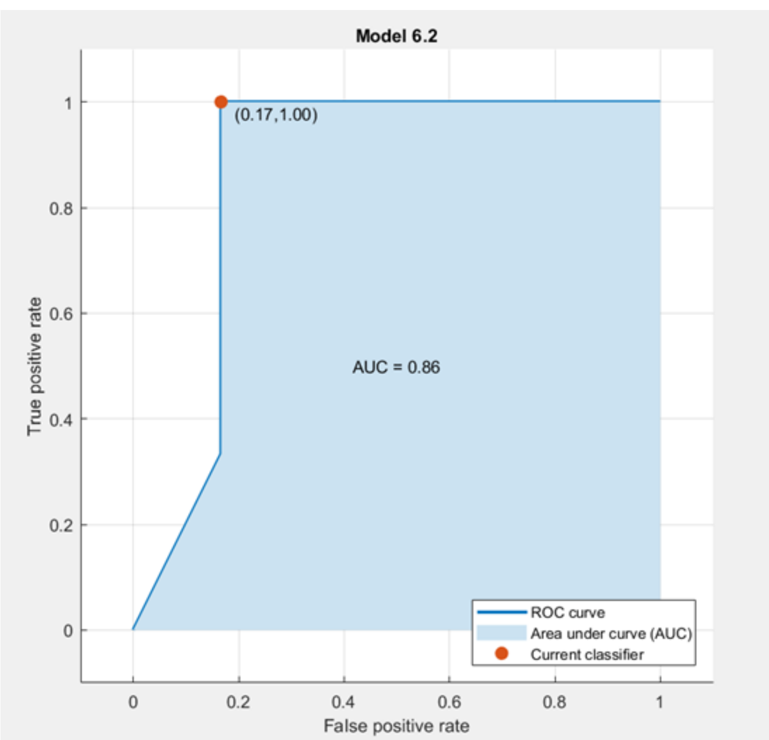
Ensembles combine multiple hypotheses to form a (hopefully) better hypothesis.

Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model, so ensembles may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation.

Fast algorithms such as decision trees are commonly used in ensemble methods (for example Random Forest), although slower algorithms can benefit from ensemble techniques as well.

An ensemble is itself a supervised learning algorithm, the trained ensemble, therefore, represents a single hypothesis. This hypothesis, however, is not necessarily contained within the hypothesis space of the models from which it is built. Thus, ensembles can be shown to have more flexibility in the functions they can represent. This flexibility can, in theory, enable them to over-fit the training data more than a single model would, but in practice, some ensemble techniques (especially bagging) tend to reduce problems related to over-fitting of the training.

Fig. 14: ROC for Ensemble: Bagged Trees model



AUC of 0.86 is peerless for our purpose

Accuracy of predicting response vector was the highest amongst all supervised classifiers at 91.7%.

4.3 Summary of performances on test batch

The numbers of patterns correctly classified are summarized in table 3. Table 4 shows the confusion matrices for Complex Tree, Quadratic Discriminant, Quadratic SVM, KNN (Fine) and Ensemble (Bagged).

Table 3: Results of the state of ripeness classification with Complex Tree, Quadratic Discriminant, QSVM, KNN(fine) and Ensemble (Bagged), in terms of pattern classified (states 1 or 0) under ripen (class 1) or unripen (class 0) and overall accuracy.

	Ripen (class 1)				Unripen (class 0)			
Berry's assigned S.L. No. →	**1**	**2**	**3**	**4**	**1**	**2**	**3**	
Method ↓								**Overall Accuracy**
Complex Tree	1	1	1	1	0	0	0	100%
Quadratic Discriminant	0	0	1	1	0	0	0	71.429%
Quadratic SVM	1	1	0	1	0	0	0	85.714%
KNN (Fine)	1	1	1	1	0	0	0	100%
Ensemble (Bagged)	1	1	1	1	0	0	0	100%

Table 4: Confusion matrices of the berry-ripeness-classification problem with Complex Tree, Quadratic Discriminant, QSVM, KNN(fine) and Ensemble (Bagged).

	Predicted Class									
	0					**1**				
True Class	C.Tree	Quad. Disc.	QSVM	KNN (fine)	Ensemble	C.Tree	Quad. Disc.	QSVM	KNN (fine)	Ensemble
0	4	3	6	6	6	2	3	0	0	0
1	3	1	1	2	1	3	5	5	4	5

5 Discussions

Since a small set of data samples was employed, a technique or classifier was needed which could still perform optimally. A 2-fold cross validation is not appropriate for small data set since we will have only half of the data set (which is already small) for training in each run. So, 'leave one out' cross validation (or at least 5 fold cross validation) would be a better option for testing. 5 fold cross validation was chosen for testing.

Which classifier is the better in case of small data samples?

From the list, decision tree is the only classification technique which does not suit small data set. Based on its greedy characteristic, small data set may cause overfitting problem.

Nearest Neighbour can in principle work with as few as one training sample; k-NN requires at least k. With linear classifier and SVM the dimension of the feature space also plays a role.

SVM is good at dealing with small data samples, since only support vectors (not all the samples) are used to construct the separating hyperplane (i.e., classifier). But SVM isn't the best option here. A common problem with this classifier is parameter tuning, which is difficult to carry out with few training samples. You're likely to obtain very unstable results.

From table 4 it can be seen that SVM would have performed better had it been subjected to parameter tuning. Complex tree model did surprisingly adept classification on small data set due to its greedy strategy; it must have been stroke of luck here.

6 Conclusion

Thermal patterns of two different sets of berries was captured as images in IR spectrum. First set served as training set for different supervised classifier models. One model from each category: Tree, Discriminant, SVM, kNN and Ensemble – was chosen based on highest level of response vector prediction accuracy amongst its category. During test phase (employing test batch), an accuracy of 100% was reached on 3 out of 5 of these; nothing astonishing considering small-scale of the data set.

However, further work to assess the longterm reliability of the system is needed. In particular, temperature difference between core and fruit surface depends on environmental temperature and so can make the method's predictions unstable. For example, washing the berries drastically reduced their temperature (both surface and core up to an extent of 5 °C) – as can be seen in Fig.1 & 4 that though few berries were present in visible spectrum image, receded from corresponding IR image as they were cooler by 5 °C after washing.

Nevertheless, we believe that a supervised machine learning classifier-based on core size and temperature relative to fruits' surface provides an attractive means of identifying the ripeness of commercial fruit.

7 Acknowledgments

The help provided by Dr. P.K. Dutta with the experimental set-up, valuable guidance and the capturing of berries' thermal images using his IR camera is gratefully acknowledged.
Satyam Awasthi also gratefully acknowledges the opportunity granted to him by Dr. P.K. Dutta, for working and exploring on the captivating subject.

References

[1] Horton B D 1992 Ripening patterns within a peach as indicated by force and soluble solids concentration J. Am. Soc. Horticultural Sci. 117 784–7

[2] Reyes M U, Paull R E, Williamson M R and Gautz L D 1996 Ripeness determination of 'Solo' papaya (Carica papaya L.) by impact force Appl. Engng. Agriculture 12 703–8

[3] Kalt W and McDonald J E 1996 Chemical composition of lowbush blueberry cultivars J. Am. Soc. Horticultural Sci. 121 142–6

[4] Vangdal E 1980 Threshold values of soluble solids in fruit determined for the fresh fruit market Acta Agriculturae Scand. 30 445–8

[5] McGlasson W B 1983 Ethylene and fruit ripening Hortscience 18 626

[6] Pare J R J and BelangerJMR (eds) 1997 Instrumental Techniques in Food Analysis (Amsterdam: Elsevier)

[7] Cho S I, Krutz G W, Gibson H G and Haghighi K 1990 Magnet console design of an NMR-based sensor to detect ripeness of fruit Trans. ASAE 33 1043–50

[8] Wai W K, Stroshine R L and Krutz G W 1995 Modified Hahn echo pulse sequence for proton magnetic resonance (1H-MR) measurements of percent soluble solids of fruits

Trans. ASAE 38 849–55 547 E Llobet et al

[9] Ward G and Nussinovitch A 1996 Peel gloss as a potential indicator of banana ripeness Lebensmittel-Wiss.-Technol. 29 289–94

[10] Galili N, Shmulevich I and Benichou N 1998 Acoustic testing of avocado for fruit ripeness evaluation Trans. ASAE 41 399–407

[11] Sharp et al., 1991; Yokoyama et al., 1991; Moffitt et al., 1992; Neven, 1994; Neven and Rehfield, 1995; Neven et al., 1996; Lurie, 1998; Mangan et al., 1998

[12] Toba and Moffitt, 1991; Neven and Mitcham, 1996; Soderstrom et al., 1996; Shellie et al., 1997

[13] Hansen, 1992

[14] S.Wang, J.Tang and R.P.Cavalieri et al., Postharvest Biology and Technology

Volume 22, Issue 3, July 2001, Pages 257-270

[15] Quinlan, J. R., (1986). Induction of Decision Trees. Machine Learning 1: 81-106, Kluwer Academic Publishers

[16] Cover TM (1965). "Geometrical and Statistical Properties of Systems of Linear Inequalities with Applications in Pattern Recognition". IEEE Transactions on Electronic Computers. EC-14 (3): 326–334. doi:10.1109/pgec.1965.264137

[17] Dagher, I. J Glob Optim (2008) 41: 15. https://doi.org/10.1007/s10898-007-9162-0

[18] Altman, N. S. (1992). "An introduction to kernel and nearest-neighbor nonparametric regression". The American Statistician. 46 (3): 175–185